

Narrowing Without Converging: How Interactive LLM Agents Search Under Feedback

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Abstract

An interactive agent searching under feedback should contract toward the answer. These agents contract and arrive nowhere. We instrument Turtle Soup, a lateral-thinking game of yes/no questions, with two process-level probes — a forced and scored checkpoint story every round and an association trajectory embedded from each round’s question — and run three Qwen tiers on 22 human-played puzzles. Interaction helps for ten of thirty rounds and then stops, at every tier, and neither the round budget, the Oracle, nor puzzle difficulty accounts for it. The per-round step contracts four to five fold over the same rounds, so the search narrows while accuracy does not move. Three explanations are tested in turn. The answer barely redirects it: controlling for round position, the smallest tier’s next move is statistically independent of whether it was just confirmed or refuted (Δ stride +0.001, 95% CI $[-0.009, +0.011]$), while the larger two move *further* after a confirmation (+0.028, +0.009), and a test that needs no premise about step size shows all three moving toward regions the Oracle has already refuted rather than toward those it has confirmed, the smallest tier most strongly. And the agent does not stop when it finds the answer: the largest tier reaches a solution-grade story in six games and volunteers one, ending with a story that sits no further from the one that scored 1.00 than any two rounds of the same game sit from each other (0.851 vs 0.841), so the moment of loss is invisible to the geometry that describes it. One endpoint number scores all of this identically. We release the harness, both instruments, the puzzle set, and every per-game log.

1 Introduction

Here is a Turtle Soup puzzle. *A man’s body lies in the desert, clutching half a matchstick; luggage and clothing are scattered around him.* That is all the solver sees. The hidden story: a hot-air balloon was losing altitude; the passengers threw out their luggage, still could not stay up, and drew lots with matchsticks; the man who drew the short one was thrown out. The solver recovers this story by asking yes/no questions, one per round.

Outcome metrics can say whether an agent got there; on our set that story is flat, and the round budget is not the constraint (§5). What they cannot say is how the agent searched: one zero-scorer never leaves its first hypothesis, another finds the answer and lets it go. What the process shows is a search that contracts without converging, and the account of why is this paper’s contribution. Turtle Soup suits the job: the behavior *is* text, the Oracle’s reply is a clean one-trit signal whose effect on the next move can be isolated, and puzzles are deliberately under-determined.

An interactive agent is supposed to *use* what it is told, so we ask first whether it does: does the answer just received change what the agent does next? Embedding each round’s question makes this measurable, and the answer is this paper’s spine, barely, and where it does, backwards. Three questions **H1**, the search narrows in on the answer as “no” answers eliminate hypotheses, so a shrinking stride means closing in. **H2**, the answer redirects the next question: a refutation should push the search away, a confirmation should refine it. **H3**, when the agent reaches the answer it will stop — the game ends at its peak, and the answer it stands on is the best one it produced. Each is what a reader reaches for first, each makes a prediction the trajectory instrument can check, and §5 takes them in turn. H1 half-holds and misleads, H2 is refuted with the sign reversed, and H3 fails at the point that matters: the largest tier reaches a solution-grade answer in six games and volunteers one.

Contributions. (1) A characterization of interactive search that **narrows without converging**: the per-round step contracts four to five fold at every tier while accuracy stays flat, and the feedback that should redirect it does not — stride conditioned on the answer received, with round position controlled, is absent at the smallest tier and inverted at the larger two, and told “no” every tier moves toward what it has already ruled out. (2) Two failure patterns **derived from it rather than catalogued**, circling and abandoning, with per-game criteria and counts. (3) An honest boundary: the geometry describes these failures but cannot score them, because the gap between holding the solution and having lost it is no larger than ordinary within-game motion. (4) Released harness, both instruments, a 22-puzzle human-played set, and every per-game log.

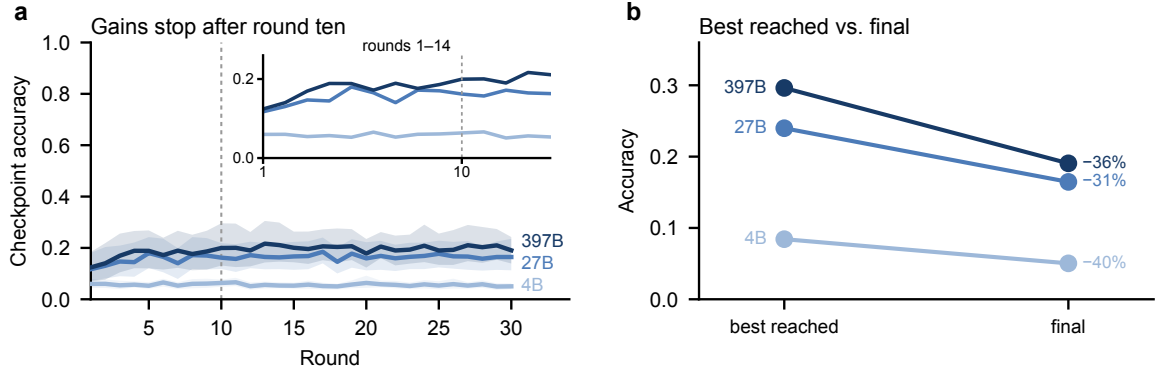


Figure 1: (a) Accuracy of the answer each tier would give at each round, on the full 0–1 range (bands, 95% cluster-bootstrap CIs over the 22 puzzles); dashed rule at round 10, inset magnifies rounds 1–14. (b) The best answer each game reached, against the answer it finally stands on.

Conditions of validity (an audited cross-family Oracle; one environment artifact removed) are stated in one sentence in §5 and specified in Appendix B; with a 95%-accurate Oracle they are no longer live problems for these results.

2 Related Work

Existing turtle-soup benchmarks, TurtleBench [1], SPLAT [2] (whose interactive player–judge protocol we adopt), and TurtleSoup-Bench [3], score verification accuracy, solve efficiency, or faithfulness to the intended story. TurtleSoup-Bench probes what agents ask next but grades it against the intended story; none of the three measures the search’s geometry, which is our delta. MUTATE [4] shows frontier agents find far fewer mechanism-distinct paths than humans; its diverge-then-narrow agent is a natural test intervention for our metrics. The instrument itself adapts forward flow [7] and the Divergent Association Task [8] from human psychometrics. The Small World of Words norms [9] provide the human anchor that ours is not yet (§3). Judge-bias catalogues [5, 6] shape the scoring: the model judge is confined to the one subscore that objective matching cannot reach.

3 Framework

3.1 Task and harness

A **Questioner** sees only the *surface* (the puzzle’s visible scenario) and asks one yes/no question per round. An **Oracle** holds the hidden story and answers yes, no, or irrelevant). The Questioner may commit a final story at any round, or is forced to at the budget. The open-source harness has pluggable model providers and logs every round, with per-game seeds.

3.2 Checkpoint and commitment instrumentation

Our first instrument does not wait for the end. Every round, after the Oracle answers, the Questioner must write its best complete story, and *that* checkpoint is scored; each point on an accuracy curve is therefore a full answer written with the evidence available at that round. Alongside it we record *commitment*: whether and when the agent ever volunteers a final story of its own accord. Together these turn an endpoint benchmark into a scored process.

3.3 Puzzles

A puzzle nobody has solved carries no evidence that its clues suffice, that its solution is unique, or that the intended leap is reachable; an agent scoring zero on it teaches us nothing. Our set therefore takes only puzzles with a public record of human play: 22 items from a Chinese Turtle Soup community site, each keeping its source URL. Surfaces run 9–108 characters (median 34), solutions 13–200 (median 102), with 97 annotated key clues. Provenance is enforced in code, with a verified family, a quarantined generated family, and tests that fail if the two mix.

Each puzzle carries an **under-determination** score: how far the surface alone leaves a solver from the solution, taken as the closest of twelve cold guesses written from the surface with no Oracle feedback. It ranges 0.173–0.577 (median 0.347) and enters §5 as a puzzle-level covariate; Appendix A gives its distribution and properties.

3.4 Outcome scoring: composite, not surface

Judged outcome is **clue recall (70)** plus **causal-logic identity (30)**; Appendix B gives the detail. String matching alone cannot tell a wrong answer from a right one worded differently: we watched a frontier model reconstruct a hidden causal chain completely and score 0.00. Conversely, the clue half blocks “far = creative” gaming, because distance without validity scores zero.

3.5 Association-trajectory instrumentation

The second instrument reads the questions themselves. Each round, we extract the question’s keywords (jieba), embed them (bge-small-zh-v1.5), and average them to a unit round vector q_t ; distances are cosine, and anchors embed the puzzle’s surface, solution, and clue terms (specification in Appendix B). Two signals follow: the **stride** $s_t = d(q_t, q_{t-1})$ and the **anchor distance** $h_t = d(q_t, \mathcal{A})$, with H1’s signature $\bar{s} \rightarrow 0$ and H2’s a rising h_t ; the lexical `question_novelty` metric is the ablation baseline that H1 predicts will miss synonym drift. One caveat up front: the anchor is a stated proxy, not a human association manifold, and its consequences are quantified with the results.

4 Experiment Designs

Designs. E1 (round curve) plays to 30 rounds under the checkpoint instrument (§3.2). E2 (round budget) imposes hard caps {5, 10, 15, 20, 25, 30} with a forced final answer only, so it doubles as a no-checkpoint control. E3 (association trajectory) computes $\{s_t\}$ and $\{h_t\}$ for every E1 game and answers H1–H3; the keyword extractor is fixed across models, so none is scored through its own vocabulary.

Feedback response, defined. For each round t we pair the Oracle’s answer (yes, no, or “irrelevant”) with s_{t+1} , the stride into the following question. A searcher that eliminates hypotheses should move *further* after a “no” than after a “yes”. Confirmation arrives earlier in a game than refutation (mean round ~ 10 vs ~ 16) and early strides are larger, so the comparison is made inside (puzzle, round-bin) cells; CIs bootstrap over puzzles. Without that control the effect is manufactured at every tier, and both estimates are reported in Appendix B.

5 Results

All numbers come from one grid on the verified set: 22 puzzles $\times$ {Qwen3.5-4B, Qwen3.6-27B, Qwen3.5-397B-A17B; hereafter 4B, 27B, 397B} $\times$ 3 seeds. **Conditions of validity, in one sentence:** the Oracle and logic judge are DeepSeek-V3.1, cross-family with the Questioners and audited at 95% on held-out probes, and one environment artifact (token exhaustion at the smallest tier) is removed in §5 before any behavioural comparison, with protocol, per-candidate audit numbers and settings in Appendix B. The three models differ in generation and architecture (the 397B is a mixture-of-experts with 17B active parameters), so we speak of model tiers, not scale; harness failures were zero, and token exhaustion is a game outcome reported below. Accuracy denotes the composite score rescaled to [0, 1]. E1 comprises 198 thirty-round games with a scored checkpoint every round; E2, 1,188 games across caps {5, 10, 15, 20, 25, 30}. Circling will emerge from the geometry (§5.3); abandoning from the checkpoint record (§5, §5.3).

What the plateau is not. Interaction helps for ten rounds and then stops (Figure 1a): the round 10 to 30 change is $-0.009 / +0.003 / -0.013$, CIs spanning zero, and each tier ends below the best answer it reached (Figure 1b). It is not the round budget: end accuracy is insensitive to the cap at the larger tiers, and the smallest model’s apparent decline is token exhaustion, rising 0/66 to 63/66 across caps with no decline among non-exhausted games. It is not a degenerate Oracle, audited at 95% with replies running 68–74% “no”. And it is not the puzzles: under-determination predicts what a model writes before any evidence arrives ($\rho = -0.34 / -0.54 / -0.47$) and nothing after it — not the round 10 to 30 change, not any trajectory indicator, not either half of the score, not the gap between tiers (all n.s. at $n = 22$, so bounds rather than absences; Appendix B). The evidence is there, the budget is there, the puzzles are within reach, and the agent stops making progress anyway.

5.1 H1: the search narrows in on the answer

It narrows; it does not converge. The intuition is that accumulated “no” answers eliminate hypotheses, so the search should contract around what survives. The contraction is real: mean stride falls monotonically at every tier, four to five fold, from 0.145 / 0.232 / 0.182 at round 1 to 0.025 / 0.053 / 0.045 at round 29. The convergence is not. Accuracy over those same rounds is flat (§5), and within a tier stride does not predict how a game ends ($|\rho| \leq 0.16$; coefficients bounded within ± 0.035 per SD, Appendix B). The narrowest searcher of all — 4B, at a late-game stride of 0.030 [0.021, 0.039] against 27B’s 0.082 [0.065, 0.098] — is the one that never reaches a solution-grade answer. Narrowing

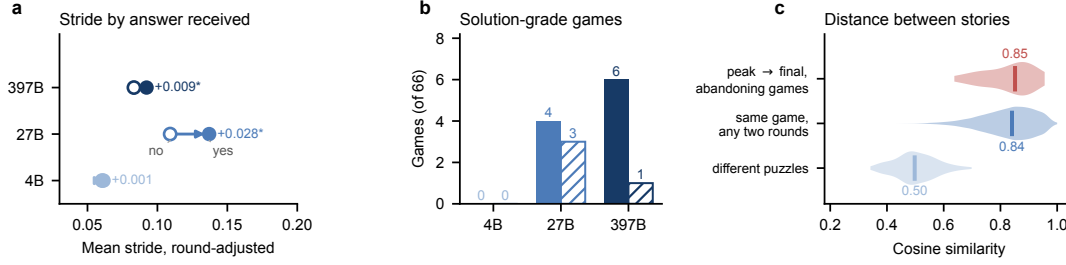


Figure 2: (a) How far the next question moves after a refutation (open marker) and after a confirmation (filled), adjusted for round position; a search that eliminated hypotheses would move further after “no”. (b) Games reaching a solution-grade answer, and how many of those the agent ever volunteered. (c) In abandoning games the story the model ends with sits as close to the story that scored 1.00 as any two moments of the same game sit to each other.

is what circling looks like from outside: orbits of one hypothesis basin, tightening around nothing. Not a malformed-question artifact: 17% of 4B’s questions are truncated or non-interrogative ($\sim 1\%$ elsewhere), yet excluding them barely moves either statistic ($0.049 \rightarrow 0.056$; $0.030 \rightarrow 0.033$).

5.2 H2: the answer redirects the next question

It barely does, and what response there is points the wrong way. Pairing each answer with the stride into the following question, inside (puzzle, round-bin) cells so that the earlier arrival of a “yes” cannot manufacture the effect (Figure 2a): 4B $+0.001$ $[-0.009, +0.011]$, no detected response; 27B $+0.028$ $[+0.010, +0.046]$ and 397B $+0.009$ $[+0.002, +0.017]$, both responding with the sign inverted. Two distinct pathologies, and neither is the search a rational agent would run.

The smallest tier does not hear the answer. Its next move is statistically independent of whether it was just confirmed or refuted, so nothing in the feedback channel can push it out of the hypothesis it started with. That is what circling is, now with a cause rather than a description.

And the direction is wrong, on a test that needs no premise about step size. Whether a refutation ought to produce a larger jump than a confirmation is not obvious: surprisal, belief change and distance in question space are different quantities, and “yes” is in fact the rare answer here (18–26% of yes/no replies), so an agent allocating steps by information content might reasonably move further after one. The following comparison sidesteps the question. For each round we ask whether the next question lands nearer to regions the Oracle has *already refuted* or to regions it has *already confirmed*. Told a “no”, all three tiers move toward what was already ruled out (-0.089 $[-0.112, -0.068]$ for 4B, -0.051 $[-0.075, -0.032]$ for 27B, -0.032 $[-0.047, -0.018]$ for 397B, puzzle-clustered). Told a “yes”, no such asymmetry appears (all three CIs span zero) — which also answers the obvious objection, since refutations outnumber confirmations three to one in both cases. Whatever the right step after a refutation is, moving back toward what has already been excluded is not it, and the smallest tier, whose stride does not respond to the answer at all, does this most strongly.

5.3 H3: when it finds the answer it will stop

With the environment artifact removed, the differences are behavioral. *It does not stop.* With H1 and H2 answered, what remains is whether the agent halts when it arrives.

It does not stop. Call a sustained peak ≥ 0.5 *solution-grade*. 4B reaches none; 27B reaches four and volunteers three; 397B reaches six and volunteers one (Figure 2b). Commitment is rare and late everywhere (11/66, 9/66, 2/66 games; mean round 11.1, 14.2, 23.5), and when it happens the agent often submits less than it had — 27B commits 0.075 on a game that peaked at 0.542. Under per-game criteria the tiers swap dominant failures: 4B circles in 39 of 66 games and abandons in 1, 397B circles in 6 and abandons in 12, with no game meeting both.

Abandoning is a sub-semantic event, and that is why the geometry cannot score it. In abandoning games the final story sits at cosine 0.851 from the peak story that scored up to 1.00, while any two checkpoints of the same game sit at 0.841 and two stories from different puzzles at 0.498 (Figure 2c). The move that loses the answer is indistinguishable, in this space, from ordinary within-game motion, though the score falls by 0.30 on average. This is a property of the phenomenon, not a defect of the instrument: a fatal substitution smaller than the encoder’s resolution cannot be predicted from that encoder.

Drift, the alternative signature, does not appear: slopes cluster near zero at every tier. In language space, lexical

novelty catches only 4B’s verbatim re-asks and cannot see abandoning at all.

And the geometry does not predict the score: within tiers no stride or drift effect is detected, every coefficient bounded within ± 0.035 per SD (Appendix B). Effects are bounded small, not excluded: the geometry says *how* an agent fails, not how well it will score.

6 Limitations and Conclusion

Limitations. Everything here comes from one game, one Oracle protocol, and three models of a single family; whether search under feedback behaves this way in other interactive settings is untested, and the title’s “interactive LLM agents” is a hypothesis about scope rather than a demonstration of one. The feedback measurement reads *how far* the next question moves, not *what* it asks; a content-level test is the natural next study, and round position is controlled only by binning. Truncated chain-of-thought under a tight token budget masquerades as an agent that cannot form a question (§5), and the Oracle’s $\sim 95\%$ audited accuracy implies roughly one wrong answer per 30-round game. The anchor is not yet human-derived (§3.5), and §5.3 shows the encoder’s resolution is itself a limit on what any embedding-based instrument can score. One DeepSeek model serves as both Oracle and judge, and the tiers confound generation and architecture, so tier effects are model comparisons, not scaling laws. The source site is public and memorisation probes are pending. Puzzles involve death and dark themes; content-flagged.

Conclusion. These agents narrow without converging. The step shrinks four to five fold while the score stands still, and the feedback that should turn the search instead sends every tier back toward what it has already ruled out — the smallest never finding the answer, the largest finding it and not stopping. The two failures need opposite fixes, and one endpoint number scores them alike.

References

- [1] Q. Yu *et al.* TurtleBench: Evaluating Top Language Models via Real-World Yes/No Puzzles. arXiv:2410.05262.
- [2] Q. Chen, B. Zhang, G. Wang, Q. Wu. Weak-eval-Strong: Evaluating and Eliciting Lateral Thinking of LLMs with Situation Puzzles. arXiv:2410.06733.
- [3] M. Zhou *et al.* What to Ask Next? Probing the Imaginative Reasoning of LLMs with TurtleSoup Puzzles. arXiv:2508.10358.
- [4] J. Park, I. Baek, J. Park, H. Lee. Beyond One Path: Evaluating and Enhancing Divergent Thinking in Interactive LLM Agents. arXiv:2605.28465.
- [5] J. Ye *et al.* Justice or Prejudice? Quantifying Biases in LLM-as-a-Judge. ICLR 2025. arXiv:2410.02736.
- [6] L. Shi *et al.* Judging the Judges: A Systematic Study of Position Bias in LLM-as-a-Judge. IJCNLP-AACL 2025.
- [7] K. Gray *et al.* “Forward Flow”: A New Measure to Quantify Free Thought and Predict Creativity. *American Psychologist* 74(5), 2019.
- [8] J. A. Olson *et al.* Naming Unrelated Words Predicts Creativity. *PNAS* 118(25), 2021.
- [9] S. De Deyne *et al.* The “Small World of Words” English Word Association Norms. *Behavior Research Methods* 51(3), 2019.

A The puzzle set

Where the puzzles come from. A puzzle nobody has solved carries no evidence that its clues suffice, that its solution is unique, or that the intended leap is reachable; an agent scoring zero on it teaches us nothing. Our set is hand-picked from the classic repertoire of a Chinese Turtle Soup community: 22 puzzles people have played and solved, each keeping its source record. Every item was read and kept or rejected by hand; rejected were puzzles turning on deduction from stated evidence rather than a lateral reframing, and puzzles whose surfaces were incomplete at the source. Surfaces run 9–108 characters (median 34), solutions 13–200 (median 102).

How hard are they? Each puzzle carries an **under-determination** score: give a model the surface alone, with no Oracle and no feedback, let it write twelve complete stories, and take the closest to the real solution. The index is one minus that best guess, and is large when the surface leaves a solver far from the answer.

Under-determination	Puzzles
0.15–0.25	5
0.25–0.35	8
0.35–0.45	4
0.45–0.55	4
0.55–0.65	1

The median is 0.347, quartiles 0.273 and 0.410, full range 0.173–0.577: single-peaked with a thin hard tail. Easiest are the story the genre is named for (0.17) and the desert matchstick (0.19). Hardest is a fourteen-character surface reading *our heights differ / a flowerpot broke / our heights are the same* (0.58).

Two properties make this usable as a covariate. It describes the puzzle, not any agent’s performance on it. It is also independent of rounds used; a difficulty defined as “how many rounds this takes” would make an accuracy-versus-rounds plot confirm itself. Nor is it a proxy for surface length ($r = -0.22$, n.s.), so length dependence was not detected at $n = 22$ (a weak test). It is one measure rather than two: a surface loose enough to admit many mechanisms is, for that same reason, one whose true mechanism takes more reframing to reach, so *how far* and *how many ways* are not separable here.

A caveat we report either way. The two puzzles scoring easiest are the two most widely circulated ones. Their surfaces may be easy to complete not because the inference is short but because the story is in the training data, in which case the measure reads familiarity as much as difficulty. A memorisation probe (asking for the solution with no surface given) separates the two and belongs in any use of this measure.

B Measuring agent performance

What counts as a correct answer. Judged outcome is **clue recall (70)** plus **causal-logic identity (30)**. Clue recall matches the answer against per-puzzle annotated key clues: objective and reproducible. The logic judge rates only whether cause $\rightarrow$ mechanism $\rightarrow$ outcome matches and is told explicitly to ignore wording. Because single ratings are unstable on borderline answers, ratings are sampled and averaged (two samples per rating in the reported grid). The gap between the two subscores is itself readable: right story, wrong vocabulary.

Clue matching tolerates paraphrase through character-bigram recall, which sets a floor on how short an annotated clue may be. Below that floor only a verbatim match counts, and the score reads vocabulary rather than content. Clues are annotated above the floor.

Does the logic half earn its thirty points? On the full grid, yes: each model earns a several-fold larger share of the logic half than of the clue half, and the halves correlate only weakly per game (Spearman $\rho = 0.30/0.35/0.06$ for 397B / 27B / 4B over 396 E2 games each). The 70/30 weighting itself is a design choice; every conclusion in §5 is readable from the two halves separately, though the absolute plateau level (~ 0.2) does depend on the weights. The logic judge rates the causal-logic half on its 30-point scale and is sampled twice per rating at temperature 0.2; the two samples agree exactly in 86% of ratings and differ by under one point of 30 on average.

Feedback response: definition, confound, and both estimates. For each round t the Oracle’s answer is paired with s_{t+1} , the stride into the next question; “irrelevant” rounds are excluded. Confirmation arrives earlier than refutation (mean round 12.1 / 9.7 / 10.9 for a “yes” against 15.6 / 15.6 / 15.9 for a “no”) and early strides are larger, so an uncontrolled comparison is confounded by position. Pairs are formed inside (puzzle, round-bin) cells with bins {1–5, 6–10, 11–15, 16–20, 21–25, 26–29}, and 95% CIs bootstrap over puzzles (4,000 resamples). Uncontrolled, all three tiers appear to respond (+0.024 [+0.004, +0.044]; +0.043 [+0.015, +0.070]; +0.034 [+0.023, +0.046]); controlled, the smallest does not (+0.001 [−0.009, +0.011]; +0.028 [+0.010, +0.046]; +0.009 [+0.002, +0.017]). The gap between the two is the point.

The semantic scale of abandoning. Within abandoning games (sustained peak ≥ 0.3 , final below half of it; $n = 34$), the final checkpoint story sits at cosine 0.851 (median) from the peak story, while any two checkpoints of the same game sit at 0.841 (median over all 77,906 within-game pairs from 60 games) and two stories from different puzzles at 0.498 (231 pairs). The score gap across that same pair is $0.447 \rightarrow 0.099$.

The round-1 prior. The round-1 checkpoint precedes the first Oracle answer and is therefore a cold guess. It holds 49% / 33% / 26% (median, by tier) of the game’s eventual best score, and correlates with it at $\rho = +0.37/+0.57/+0.47$ (permutation $p = 0.09/0.003/0.03$, $n = 22$ puzzles with seeds averaged). Under-determination, itself measured from twelve cold guesses by a model cross-family with every Questioner, predicts round-1 quality at $\rho = -0.34/-0.54/-0.47$ ($p = 0.12/0.010/0.029$) but not the final score.

Behavioural indicators against puzzle structure. Regressing each per-trace indicator on under-determination, seeds averaged, 5,000 permutations: stride, late stride, drift (both anchors), mean anchor distance and rounds played are all n.s. at every tier ($|\rho| \leq 0.32$, $p > 0.10$). Commit rate is negative at all three ($-0.32/-0.25/-0.43$) but pooled $p = 0.13$, and 1 of 24 tests reaching $p < 0.05$ is the chance expectation, so we report it as unresolved rather than as an effect.

The two score halves dissociate (a check on the scoring instrument, moved here from the body). Every model earns proportionally far more of the judged causal-logic half than the clue-recall half: 397B averages 12.6/30 logic against 7.2/70 clues, 27B 10.1 against 6.8, and 4B 2.9 against 0.2, and the halves correlate only weakly per game (Spearman $\rho = 0.30/0.35/0.06$). Models reach the right causal shape well before, and often without, landing the annotated clue content.

Retention estimator and permutation null. Retention is the mean per-game final-to-sustained-peak ratio over games with a nonzero sustained peak; the final is the last played checkpoint, and excluding early-committing games changes each tier’s value by ≤ 0.03 (bootstrap 95% CIs $\pm \approx 0.08$). For the null: unit, the game. For each of 2,000 permutations, every game’s checkpoint sequence is shuffled within the game (knowledge-constant, jitter-only null), the sustained peak and the final (last element) are recomputed on the shuffled sequence, and retention is the mean over that tier’s games with a nonzero permuted peak. The reported p is one-sided per tier, the fraction of permutations whose null retention is at or below the observed value. Late-game stride means the second half of a game’s strides.

Mixed-model coefficients. Best checkpoint accuracy per game; Gaussian LMM, puzzle random intercept, tier fixed effect, ML fits, likelihood-ratio tests against the covariate-only model. Stride $\beta = -0.010$ per SD, 95% CI $[-0.035, 0.014]$, $p = 0.42$; drift $\beta = -0.011$, $[-0.035, 0.014]$, $p = 0.39$; stride $\times$ tier $p = 0.20$; surface-anchor drift $\beta = -0.020$, $[-0.044, 0.003]$, $p = 0.097$.

Environment settings and Oracle audit. Every model call uses max_tokens 2048 (thinking disabled, temperature 0.2) with a per-game budget of 50,000 tokens; exhausting the game budget forces a final answer and is recorded as the game outcome token_budget. The Oracle audit probes a candidate with 25 hand-crafted (question, expected-answer) pairs over three puzzles, 21 with a definite yes/no answer and 4 expecting “irrelevant”; the pass bar is 90% on the yes/no items. The deployed DeepSeek-V3.1 scored 20/21 (95%) on yes/no and 3/4 exact on “irrelevant”. Candidates measured in the pilot phase: a 4B Oracle with thinking disabled, 30% (rejected; this is the configuration behind abstract finding 4, where raising Oracle accuracy from 30% to 100% left the 4B Questioner’s score unchanged); the same model with thinking enabled, 70% (rejected); a same-family flash model, 90% (excluded for sharing the Questioners’ family); GLM-5.3-flash, 100% (used in earlier tooling).

Trajectory and difficulty pipeline. Keywords are extracted with jieba after stripping Latin characters and digits; each keyword is embedded with bge-small-zh-v1.5 (normalized), and the round vector is the re-normalized mean of its keyword embeddings. All distances are cosine. The solution-aware anchor embeds the puzzle’s surface, solution, and clue terms; the surface-only anchor embeds surface terms alone. The under-determination measure prompts GLM-5.3-flash (cross-family with all Questioners) for twelve cold guesses from the surface and scores closeness as embedding similarity between guess and solution under the same encoder, not with the composite score, so the covariate is not coupled to the outcome apparatus.

Artifacts. Harness, puzzle set with per-item source records, per-game logs (including every checkpoint story), and the analysis scripts behind each reported number are released at github.com/RainyFields/Project_Turtle_Soup. Puzzle surfaces and solutions originate from a public community repertoire and are redistributed with source attribution and a content flag.

Where the accuracy curve comes from. E1 does not score once at the end. Every round, after the Oracle answers, the Questioner is required to commit its best complete story, and *that* is scored. Each point on the accuracy curve is therefore a full answer written with the evidence available at that round.

One game, worked. Puzzle refsoup_021 (surface: a stormy night, a run-down temple, two photographs on the

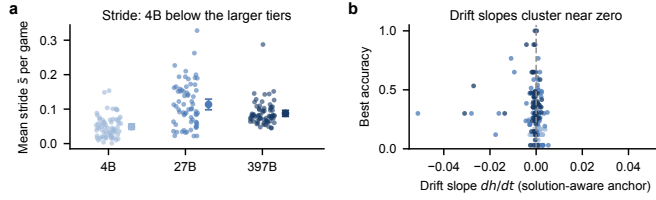


Figure 3: Per-trace geometry ($n = 198$). **(a)** Per-game mean stride by tier (dots; bars, tier mean $\pm$ 95% CI): 4B sits below the larger tiers. **(b)** Drift slope against best accuracy, clustered near zero at every tier.

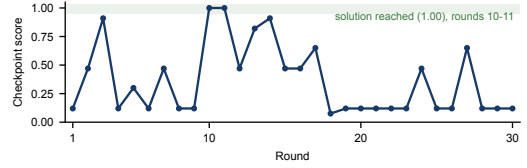


Figure 4: One abandoning game (refsoup_021, 397B, seed 2): the round 10–11 checkpoint stories are the solution (band), never volunteered, abandoned by round 19. Full rows in Appendix B.

wall that seem to stare; by morning the narrator sees only two windows; solution: “there were never photos on the wall; people were pressing against the windows, looking in”), played by Qwen3.5-397B, seed 2. Table 1 shows seven of thirty rounds; Questions and stories are our translations of the logged Chinese, stories abridged to the clause carrying the hypothesis. Rounds 22–30, eight lexically distinct re-asks of the same flooding hypothesis (e.g. round 25, whether the narrator woke floating upside down; round 28, whether rising water reached eye level), are in the released logs.

Rd	Question (abridged)	Ans.	Committed story (abridged)	score (clue+logic)
1	Are the ones in the photos people?	yes	The two “photos” are two people nailed or sealed into wall niches.	0.12 (0+12)
3	On waking the next day, is he still in the temple?	yes	The “two photos” are really two windows, mistaken for photos in the dim light.	0.91 (70+21)
4	Are the “two windows” the same objects as last night’s “photos”?	yes	The “photos” were hollowed-out window openings; the fear came from darkness and illusion.	0.12 (0+12)
10	Did frame glass break and fall, leaving open holes?	no	They were never photos, but faces pressed against the outside of the windows.	1.00 (70+30)
11	Were the photos removed, revealing windows behind?	no	The staring figures were people outside the windows.	1.00 (70+30)
19	Were real people hiding behind the photos?	no	Window openings sealed by painted cloth covers; the storm tore them off.	0.12 (0+12)
30	Did rising water lift the narrator higher?	no	Papered-over window openings; the storm stripped the paper away.	0.12 (0+12)

Table 1: Seven rounds of one E1 game (refsoup_021, Qwen3.5-397B, seed 2): the Questioner’s question, the Oracle’s answer, the story the agent was forced to commit that round, and its composite score. Italics are our translations; the original transcripts are in the released logs.

Three things the table shows that no mean curve can. **The checkpoint mechanism surfaces knowledge the agent never acts on.** The round-10 committed story *is* the solution, synthesized from earlier answers on a round whose own question was answered “no”. Yet the model never volunteers a final answer in all thirty rounds (one of the 57/66 non-committing 397B games), and by round 19 it has abandoned the correct reading for a falling-coverings mechanism it never escapes. **Stalling in fresh words happens at the largest scale too.** Rounds 22–30 are eight lexically distinct re-asks of the same flooding hypothesis, exactly the pattern lexical novelty misses. **The curve’s round-to-round jitter is a scoring fact, not a knowledge fact.** The same understanding re-worded (rounds 3 $\rightarrow$ 4) swings the clue half 70 $\rightarrow$ 0. The mean curves of §5 average this out, which is why single-game curves are not the unit of analysis.